WORKED SOLUTION

STEP III 1995, Question 6

Pure • Loci and Inversion

STEP III 1995, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q6

The key idea

The circle $|z - \mathrm{i}| = 1$ passes through the origin, so in polar form its equation is especially simple: the modulus is a single trigonometric function of the argument. Once we have z in modulus–argument form, $1/z$ is immediate (take the reciprocal of the modulus and negate the argument), and writing it in Cartesian form reveals that the circle has become a straight line. The final part is the same phenomenon run in reverse, and it is quickest to do it directly with $w = x + \mathrm{i}y$.

Solution

The modulus of z

Write $z = r(\cos\theta + \mathrm{i}\sin\theta)$ with $r > 0$. Then $z - \mathrm{i} = r\cos\theta + \mathrm{i}(r\sin\theta - 1)$, so

$$\begin{aligned} |z - \mathrm{i}|^2 = 1 &\iff r^2\cos^2\theta + r^2\sin^2\theta - 2r\sin\theta + 1 = 1 \\ &\iff r^2 - 2r\sin\theta = 0 \\ &\iff r = 2\sin\theta, \end{aligned}$$

where in the last step we divided by r, which is allowed because $z \neq 0$. So

$$|z| = 2\sin\theta.$$

Since $r > 0$, this forces $\sin\theta > 0$, so $0 < \theta < \pi$: the circle (centre i, radius 1) lies in the upper half-plane and touches the real axis only at the origin, which is excluded. Conversely, every $\theta \in (0, \pi)$ gives a point $z = 2\sin\theta\,(\cos\theta + \mathrm{i}\sin\theta)$ of the circle, so as θ runs through $(0, \pi)$, z traces the whole circle except the origin.

Ask yourself Why is $|z| = 2\sin\theta$ "obvious" from a picture?

The segment from 0 to 2i is a diameter of the circle. By the angle in a semicircle, the triangle with vertices 0, z, 2i has a right angle at z, and the angle at 0 is $\frac{\pi}{2} - \theta$. So $|z| = 2\cos(\frac{\pi}{2} - \theta) = 2\sin\theta$.

The modulus and argument of $1/z$

Since $z = 2\sin\theta\,(\cos\theta + \mathrm{i}\sin\theta)$,

$$\frac{1}{z} = \frac{1}{2\sin\theta}(\cos(-\theta) + \mathrm{i}\sin(-\theta)),$$

so

$$\left|\frac{1}{z}\right| = \frac{1}{2\sin\theta}, \qquad \arg\frac{1}{z} = -\theta.$$

Because $0 < \theta < \pi$, the angle $-\theta$ lies in $(-\pi, 0)$, so it is the principal argument.

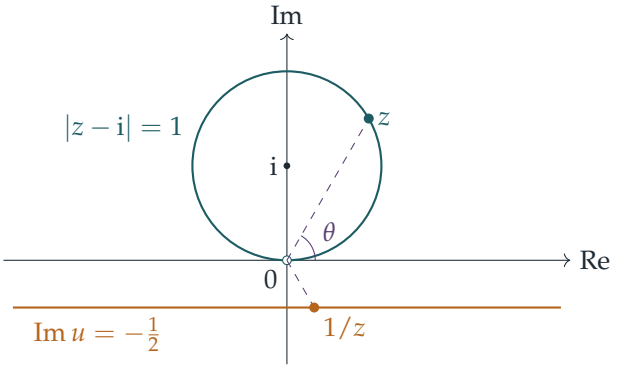
The loci

To see what $1/z$ traces, convert to Cartesian form:

$$\frac{1}{z} = \frac{\cos\theta - \mathrm{i}\sin\theta}{2\sin\theta} = \tfrac{1}{2}\cot\theta - \tfrac{1}{2}\mathrm{i}.$$

The imaginary part is always $-\frac{1}{2}$, and as θ runs through $(0, \pi)$ the real part $\frac{1}{2}\cot\theta$ takes every real value exactly once (it decreases from $+\infty$ to $-\infty$). So $1/z$ traces the *whole* straight line $\operatorname{Im} u = -\frac{1}{2}$.

So z lies on the circle with centre i and radius 1 (minus the origin), and $1/z$ lies on the horizontal line through $-\frac{1}{2}\mathrm{i}$. The diagram shows both, with the pair of points for $\theta = \frac{\pi}{3}$: here $z = \frac{\sqrt{3}}{2} + \frac{3}{2}\mathrm{i}$ and $1/z = \frac{1}{2\sqrt{3}} - \frac{1}{2}\mathrm{i}$. The two points lie on rays at angles θ and $-\theta$, and their distances from 0 multiply to 1. As z approaches the origin, $1/z$ runs off to infinity along the line; the top of the circle, $z = 2\mathrm{i}$, goes to the point $-\frac{1}{2}\mathrm{i}$, the nearest point of the line to the origin.



The locus C

Let $w = x + \mathrm{i}y \neq 0$ (we need $w \neq 0$ for $1/w$ to exist). Then

$$\frac{1}{w} = \frac{\bar{w}}{|w|^2} = \frac{x - \mathrm{i}y}{x^2 + y^2}, \qquad \text{so} \qquad \operatorname{Re}\frac{1}{w} = \frac{x}{x^2 + y^2}.$$

Hence, for $w \neq 0$,

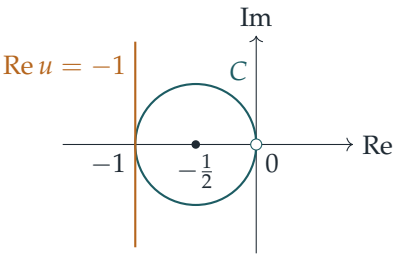
$$\begin{aligned} \operatorname{Re}\frac{1}{w} = -1 &\iff x = -(x^2+y^2) \\ &\iff x^2 + x + y^2 = 0 \\ &\iff (x+\tfrac{1}{2})^2 + y^2 = \tfrac{1}{4}. \end{aligned}$$

Every step is reversible (we multiplied by x^2+y^2, which is non-zero), so this is a genuine "if and only if". Therefore C is the circle

$$|w + \tfrac{1}{2}| = \tfrac{1}{2},$$

with centre $-\frac{1}{2}$ and radius $\frac{1}{2}$, *with the origin removed*. The origin does lie on this circle, but $1/w$ is undefined there, so it is not in C.

This mirrors the first part: there a circle through 0 was sent by $z \mapsto 1/z$ to a line not through 0; here the line $\operatorname{Re} u = -1$ is sent by the same map (which is its own inverse) to a circle through 0.



Checks

- Take $\theta = \frac{\pi}{6}$: then $|z| = 1$ and $z = \frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i}$, and indeed $|z - \mathrm{i}| = |\frac{\sqrt{3}}{2} - \frac{1}{2}\mathrm{i}| = 1$. Its reciprocal is its conjugate, $\frac{\sqrt{3}}{2} - \frac{1}{2}\mathrm{i} \approx 0.866 - 0.5\mathrm{i}$, which has imaginary part $-\frac{1}{2}$ as predicted.
- $\theta = \frac{\pi}{2}$ gives $z = 2\mathrm{i}$, the top of the circle, and $1/z = -\frac{1}{2}\mathrm{i}$.
- For C: $w = -1$ is on the circle and $1/w = -1$; $w = -\frac{1}{2} + \frac{1}{2}\mathrm{i}$ is on the circle and $1/w = -1 - \mathrm{i}$, with real part -1.

A numerical test of 10 000 random points on each locus confirmed all three results.

Common slips

- Forgetting the range $0 < \theta < \pi$: the formula $|z| = 2\sin\theta$ only makes sense where $\sin\theta > 0$.
- Describing the locus of $1/z$ as a half-line or segment. Since $\cot\theta$ takes all real values on $(0, \pi)$, it is the whole line.

- Including the origin in C. It satisfies the circle's equation but $1/w$ does not exist there.

Beyond STEP

The map $z \mapsto 1/z$ is an example of a Möbius transformation, $z \mapsto (az+b)/(cz+d)$. A central fact about these maps is that they send "generalised circles" (circles and straight lines) to generalised circles, and the circles through the origin are exactly the ones that $1/z$ sends to lines. This is the beginning of the geometry of the Riemann sphere, on which a line is just a circle through the point at infinity. Needham's *Visual Complex Analysis* develops all of this beautifully, with pictures.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.